LEVEL UP ACADEMY

Class 10 | GSEB | Science

Ch 12 Test (GSEB pattern): Magnetic Effects of Electric Current

Chapter / Topic: Magnetic Effects of Electric Current

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: objective questions, 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 4 marks.
3. Write balanced equations and draw neat, labelled diagrams wherever needed.

SECTION A

9 marks

Choose the correct option.

Q1. Which of the following correctly describes the magnetic field near a long, straight, current-carrying wire? **[1]**

(a) Straight lines perpendicular to the wire
(b) Straight lines parallel to the wire
(c) Concentric circles centred on the wire
(d) Radial lines starting from the wire

Q2. Outside a bar magnet, magnetic field lines go: **[1]**

(a) from the north pole to the south pole
(b) from the south pole to the north pole
(c) in straight lines away from both poles
(d) only near the middle of the magnet

Q3. The magnetic field inside a long, straight, current-carrying solenoid: **[1]**

(a) is zero
(b) decreases towards its ends
(c) increases towards its ends
(d) is the same at all points

Q4. A current-carrying conductor placed parallel to a magnetic field experiences: **[1]**

(a) the maximum force
(b) no force
(c) a force along the field
(d) a force opposite to the field

Q5. A positively charged particle (alpha particle) moving towards the west is deflected towards the north by a magnetic field. The direction of the magnetic field is: **[1]**

(a) towards the south
(b) towards the east
(c) upward
(d) downward

Q6. Which instrument detects the presence of a current in a circuit? **[1]**

(a) Galvanometer
(b) Voltmeter
(c) Rheostat
(d) Solenoid

Q7. Magnetic field lines: **[1]**

(a) can intersect each other
(b) never intersect each other
(c) intersect only at the poles
(d) intersect at neutral points

Q8. Which of the following increases the strength of an electromagnet? **[1]**

(a) Reducing the current
(b) Reducing the number of turns
(c) Using a wooden core
(d) Using a soft iron core

Q9. Which device protects a household circuit from short-circuiting and overloading? **[1]**

(a) Fuse (b) Ammeter
(c) Voltmeter (d) Rheostat

SECTION B

6 marks

Q10. State Fleming's left-hand rule. **[2]**

Q11. Describe the magnetic field lines of a current-carrying circular loop. Why is the field strong at the centre? **[2]**

Q12. What is the function of the earth wire? Why should metallic appliances be earthed? **[2]**

SECTION C

6 marks

Q13. Answer the following about a solenoid: **[3]**

(i) What is a solenoid? **[1]**

(ii) How does the magnetic field of a current-carrying solenoid compare with that of a bar magnet? **[1]**

(iii) How can a solenoid be used to make an electromagnet? **[1]**

Q14. A circular loop of wire lies flat on a table, and current flows through it clockwise (as seen from above). **[3]**

(i) State the right-hand thumb rule. **[1]**

(ii) What is the direction of the magnetic field inside the loop? **[1]**

(iii) What is the direction of the magnetic field outside the loop? **[1]**

SECTION D

4 marks

Read the passage and answer the questions that follow.

Q15. Rohan's house has two separate circuits: one rated 15 A for high-power appliances like the geyser and AC, and one rated 5 A for fans and bulbs. One day, an electric heater, an iron and the AC were all switched on together from the same circuit, and the MCB tripped. **[4]**

(i) Why are separate circuits with different current ratings used? **[1]**

(ii) Why did the MCB trip? **[1]**

(iii) Calculate the current drawn by a 2 kW heater on a 220 V supply. Can it be used safely on the 5 A circuit? **[2]**

~ All the best! ~